Lithuanian ablaut as floating features: a case against readjustment

Veronika Gvozdovaitė veronika.g@nyu.edu **New York University**

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Treating phonological changes in the root via a special mechanism (readjustment rules) cannot capture Lithuanian root patterns.

Takeaways

- Some phonological changes happen in a specific morphosyntactic environment
  English past tense: *give – gave, come – came, …*
- An approach dating back to the Sound Pattern of English (Chomsky & Halle 1968) and attributed to Distributed Morphology: (Halle & Marantz 1993)

(1) *Readjustment rule*
Nucleus(Y) → /eɪ/ / __ PAST , where Y = {give, come, …}
(Nucleus(Y): applies to a root; / __ PAST: in a morphosyntactic context)

- Rules of this format are unable to capture the Lithuanian pattern of morphosyntactically-conditioned root phonology
- But a small change in rule format makes all the difference: VI rules don't have the same problem

(2) *Vocabulary Insertion rule*
PAST ↔ [F] / __ {give, come …}
(PAST: applies to a morphosyntactic feature; / __ {give, come …}: in the context of a root)

0. Vowels in the Lithuanian verb

- Lithuanian roots show vowel changes (= ablaut)

Ablaut in three environments.

šɪl-ti šɪl-o šɪl-s	'warm-INF' 'warm-3.PAST' 'warm-FUT'	ši:l-a	'warm-3.**PRES**'	length
gɪn-ti gɪn-a gɪn-s	'defend-INF' 'defend-3.PRES' 'defend-FUT'	gi:n-e	'defend-3.**PAST**'	length
lɪnk-ti	'get bent' (= bend.INCH)	lenk-ti	'bend' (= bend.**CAUSATIVE**)	height

- The same root participates in several of these: 4 forms total for 'rise'

	'rise'	'rise.**CAUSATIVE**'
Present	ki:l	kel
Past	kɪl	ke:l
Future	kɪl	kel

Generalizations:
(i) For a set of roots, a tense feature (PAST / PRESENT) is associated with a long vowel in the root.
(ii) For a set of roots, a causative form is associated with a lower vowel than an inchoative form.

2. Readjustment fails to generate the pattern

- Readjustment rules like (1) are phonological:
  - apply after spellout
  - apply to roots in a morphosyntactic context
- A more complex pattern (Lithuanian) runs into several problems

(3) Readjustment rules that capture the generalization
Nucleus(Y1) → [+long] / __ PRESENT, where Y1 = {rise, warm, …}
Nucleus(Y2) → [+long] / __ PAST, where Y2 = {rise, defend, …}
Nucleus(Y3) → [–high] / __ CAUS, where Y3 = {rise, bend, …}

√rise lengthens in the present

√rise lengthens in the past

	'rise'	'rise.**CAUSATIVE**'
Present	ki:l	ke:l ☹
Past	ki:l ☹	ke:l
Future	kɪl	kel

- PAST/PRESENT lengthens, CAUS lowers. ✓Generalization captured

! This **overgenerates.** (see table above)

(4) Readjustment fix: enriched context
Nucleus(Y2) → [+long] / __ **PAST, CAUS** where Y1 = {rise, …}

! (4) still **overgenerates**: CAUS doesn't block PRES-ablaut. (see table below)

Possible fixes:
- Nucleus(Y1) → [+long] / __ **PRES, INCH** is unorthodox syntactically (INCH is 'unmarked')
- Locality condition: CAUS intervenes between √rise and PRESENT? No: √rise-PAST interaction happens even over CAUS

	'rise'	'rise.**CAUSATIVE**'
Present	ki:l	ke:l ☹
Past	kɪl	ke:l
Future	kɪl	kel

√rise still lengthens in the present

! (4) **misses the generalization:** lengthening is associated with tense, and has nothing to do with the causative; but in (4), they are equal triggers

! (4) is **redundant**: we still need the PAST-only rule (2) for 'defend': non-CAUS roots that still undergo past-tense lengthening

Readjustment rules (structural description = root) cannot capture the Lithuanian pattern where **a single root** has 4 phonological forms.

3. Bonus: several ablaut types

- PAST-tense ablaut has two types: some roots lengthen, some raise

+long ablaut			–low ablaut		
gɪn-a	'defend-3.PRES'	gi:n-e 'defend-3.**PAST**'	kɑ:r-ɟa	'hang-3.PRES'	kɔ:r-e 'hang-3.**PAST**'

- We need to list which roots the ablaut applies to (=lexical index): not phonologically general (compare the two columns)

The rime of the root does not predict whether ablaut occurs.

ablauting			*non-ablauting*		
root	*PAST*	*gloss*	*root*	*PAST*	*gloss*
vem	**ve**:m	'vomit'	tem	tem	'darken'
gɪn	**gi**:n	'defend'	gɪn	gɪn	'shepherd'
kul	**ku**:l	'whack'	gul	gul	'lay'
kɑu	**kɔ**:v	'fight'	kau	kav	'go'
kɑ:r	**kɔ**:r	'hang'	bɑ:r	bɑ:r	'reprimand'

Two types of PAST ablaut.

root	*PAST*	*gloss*	*count*	
lengthening ablaut				
kel	**ke**:l	'raise'	23	[–low] vowels
gɪn	**gi**:n	'defend'	36	
kul	**ku**:l	'whack'	26	
raising ablaut				
ʃɑu	**ʃɔ**:v	'shoot'	15	[+low] vowels
kɑ:r	**kɔ**:r	'hang'	1	

- But once we know that a root is ablauting, the type of ablaut is phonologically predictable:
  - Non-low vowels lengthen,
  - Low vowels raise (see table)
- Unified under one VI rule:

(5) PAST ↔ { [+long], [–low] } / __ {{rise, CAUS}, defend, hang, …}

- Both floating features dock: one is vacuous (machinery inspired by Wolf 2007)
  - The output is always [+long] and [–low]
  - –low remains –low, becomes +long
  - +low becomes –low, remains +long

Both floating features dock: lengthening.

	gɪn + [+long]$_3$ [–low]$_4$	MAXFLT	IDENT
	[–long$_1$, –low$_2$]		
☞ a.	gi:n [+long$_3$, –low$_4$]		**
b.	gi:n [+long$_3$, –low$_2$]	*!	*
c.	gɪn [–long$_1$, –low$_2$]	**!	*

Both floating features dock: raising.

	kɑ:r + [+long]$_3$ [–low]$_4$	MAXFLT	IDENT
	[+long$_1$, +low$_2$]		
☞ a.	kɔ:r [+long$_3$, –low$_4$]		**
b.	kɔ:r [+long$_1$, –low$_4$]	*!	*
c.	kɑ:r [+long$_1$, +low$_2$]	**!	*

2. Vocabulary Insertion of [F]

- Vocabulary Insertion:
  - inserts phonological material (syntax → PF)
  - applies to any node (not just roots)
- No problems if PAST, PRESENT, CAUSATIVE are insertion targets rather than contexts
  - The contexts are roots or spans (Merchant 2015, Svenonius 2012)
  (compare (3) & (5))
- ✓ captures the generalization that PAST is associated with [+long], CAUSATIVE with [–high]

When the target of phonological insertion is the syntactic feature, the pattern is easily captured.

(5) Vocabulary Insertion rules
PRESENT ↔ [+long] / __ {rise, warm, …}
PAST ↔ [+long] / __ {{rise, CAUS}, defend, …}
CAUS ↔ [–high] / __ {rise, bend, …}

{√rise, CAUS} is the relevant piece

× √rise-PRES are non-local: regular assumption for VI

✓ {√rise, CAUS} is the relevant piece: local to tense

	'rise'	'rise.**CAUSATIVE**'
Present	ki:l	kel
Past	kɪl	ke:l
Future	kɪl	kel

Conclusions

- Lithuanian root vowel alternations (= morphosyntactically-conditioned phonological changes on the root) can be captured by viewing them as item-and-arrangement morphology, with floating features as items
- A minimally different phonological approach (readjustment rules) is in principle unable to capture these data due to the format of the rule

Selected references: Chomsky, N. & Halle, M. 1968. *The Sound Pattern of English*. Halle, M. & Marantz, A. 1993. Distributed Morphology and the Pieces of Inflection. In *The View from Building 20*. MIT Press. Merchant, J. 2015. How much context is enough? Two cases of span-conditioned stem allomorphy. *Linguistic Inquiry*. Svenonius, P. 2012. Spanning. Ms., University of Tromsø. Wolf, M. 2007. For an autosegmental theory of mutation. In *UMOP 32: Papers in Optimality Theory III*.